SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	CSE Ai	**Date:**	26-08-2026

Code of Conduct

I L Sai Pavan_ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: L Sai Pavan

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason

d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. **Maintain entity coherence using assignment, mistakes, and you.**
2. **Use a Cause–Effect or Elaboration discourse relation.**
3. **Include at least two keywords from:**
   - **review**
   - **improve**
   - **confident**
   - **feedback**
4. **Response length must be 2–3 sentences.**

5. **Maintain a positive and polite tone.**
6. **Do not give contradictory advice.**

Tasks:

a) Identify the dialog acts required.

b) Identify the important entities that must be maintained.

c) Generate three possible responses satisfying all constraints.

d) Compare the responses based on:

- **Coherence**
- **Constraint satisfaction**
- **Politeness**
- **Relevance**

e) Select the best response and justify your answer.

f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- **A financial institution**
- **The side of a river**

Constraints:

1. **Resolve the correct word sense using contextual information.**
2. **Identify important entities and actions.**
3. **Represent the meaning using predicate logic.**
4. **Preserve semantic relationships.**
5. **Maintain coherence in the generated paraphrase.**

Tasks:

a) Identify the possible meanings of bank.

b) Select the correct sense and justify your decision using contextual constraints.

c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Assignment: Reference Resolution, Dialog Acts, and Word Sense Disambiguation

Introduction

Natural Language Processing systems need to understand relationships between words and sentences in order to interpret human language correctly. Reference resolution helps identify what pronouns and referring expressions refer to, while discourse coherence helps maintain meaning across sentences. Dialog act recognition allows a system to identify the purpose of a user's statement and generate an appropriate response. Word Sense Disambiguation helps determine the intended meaning of an ambiguous word using its surrounding context.

This assignment analyses reference resolution, dialog act recognition, and word sense disambiguation using the given examples.

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1. Reference Resolution and Discourse Coherence

Given Paragraph

“Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving.”

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a) Referring Expressions and Possible Antecedents

A referring expression is a word or phrase that refers to an entity already mentioned or understood in the discourse.

The important referring expressions are:

1. “He”

Possible antecedents:

- Ravi
- Arun

The pronoun is singular and masculine, so both Ravi and Arun satisfy gender and number agreement.

However, recency and grammatical role suggest that **Ravi** is the more likely antecedent.

2. “him”

Possible antecedents:

- Ravi
- Arun

Both are singular and masculine. However, the surrounding discourse suggests that **Arun helped Ravi**, making Ravi the likely antecedent.

3. “it”

Possible antecedents:

- Book
- Artificial Intelligence

The pronoun **it** is singular and can refer to the book. It is semantically compatible with **find it**, because a person can find a book.

Therefore, **it = the book**.

4. “they”

Possible antecedents:

- Ravi and Arun

The pronoun **they** is plural. Since Ravi and Arun are the two people introduced in the paragraph, **they** refers to both of them.

5. “the book”

This is a definite referring expression referring to the book introduced earlier.

6. “leaving”

The understood subjects of **leaving** are Ravi and Arun because they are the participants who discussed the book before leaving.

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b) Applying Reference Resolution Constraints

Gender and Number Agreement

Both Ravi and Arun are male and singular.

Therefore:

- **He → Ravi or Arun**
- **him → Ravi or Arun**

Both candidates satisfy gender and number agreement.

For **they**, the antecedent must be plural. Ravi and Arun together form a plural group.

For **it**, the antecedent must normally be singular and semantically appropriate. **Book** satisfies this requirement.

Recency

Recency generally gives preference to recently mentioned compatible entities.

In the first sentence, both Ravi and Arun are introduced. Arun is mentioned immediately before the pronoun **He**.

Therefore, recency alone could favour **Arun**.

However, recency is not sufficient by itself because other discourse constraints must also be considered.

Grammatical Role

The subject of the first sentence is **Ravi**:

Ravi met Arun.

The next sentence begins with:

He was looking for a book.

Maintaining the subject/topic of the previous sentence makes **Ravi** a strong candidate.

In the third sentence:

Arun helped him find it.

Arun is explicitly named as the subject, while **him** is the object. This makes **Ravi** the natural interpretation of **him** because the sentence establishes Arun as the helper and Ravi as the person being helped.

Semantic Compatibility

The phrase:

"was looking for a book"

is naturally applicable to a person.

Both Ravi and Arun can logically look for a book.

The phrase:

"helped him find it"

also supports the interpretation that Arun helped Ravi find the book.

The pronoun **it** is most compatible with **book**, because a book is something that can be found.

Discourse Coherence

The paragraph presents a coherent sequence:

1. Ravi and Arun meet.
2. Ravi looks for a book.
3. Arun helps Ravi find the book.
4. Ravi and Arun discuss the book.
5. They leave.

This interpretation creates a consistent discourse in which both people participate in the events.

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c) Reference Resolution Table

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
He	Ravi	**Valid**	Singular, male, grammatical subject/topic, and coherent with later sentence
He	Arun	Possible but less preferred	Gender and number agree, but discourse continuity favours Ravi
him	Ravi	**Valid**	Singular male and semantically fits the role of person being helped
him	Arun	Less preferred	Arun is already the explicit subject/helper in the sentence
it	Book	**Valid**	Singular and semantically compatible with “find”
it	Artificial Intelligence	Invalid/less suitable	“Find Artificial Intelligence” does not fit the immediate discourse meaning
they	Ravi + Arun	**Valid**	Plural pronoun referring to both people
they	Ravi alone	Invalid	Singular entity cannot match plural pronoun
they	Arun alone	Invalid	Singular entity cannot match plural pronoun
the book	Previously mentioned book	**Valid**	Definite noun phrase refers back to the introduced book

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d) Final Coreference Chains

The final coreference chains are:

Person 1

Ravi → He → him

These expressions refer to Ravi.

Person 2

Arun → Arun

Arun is explicitly named in both relevant sentences.

Book

a book → it → the book

These expressions refer to the same book.

Group

Ravi + Arun → they

The plural pronoun **they** refers to Ravi and Arun together.

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e) Paragraph with Pronouns Replaced

The paragraph can be rewritten as:

“Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before Ravi and Arun left.”

This version removes the pronouns and makes the intended references explicit.

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f) Ambiguity if “He Was Looking for a Book” Appeared Before Arun Was Introduced

If the paragraph were structured as:

“Ravi met at the library. He was looking for a book. Arun arrived later.”

then **He** would have only one previously introduced male antecedent, Ravi.

However, if the sentence were presented before Ravi had been introduced, there would be no clear antecedent for **He**.

For example:

“He was looking for a book. Ravi met Arun at the library.”

The pronoun **He** appears before Ravi and Arun are introduced. This creates a **forward-reference or discourse ambiguity** because the reader does not yet know who **He** refers to.

A good NLP system should therefore consider discourse context and not rely only on local grammatical information.

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2. Dialog Act Recognition and Response Generation

Given Conversation

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

The required dialog acts are:

Reassure + Advise

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a) Required Dialog Acts

There are two required dialog acts.

Reassure

The response should reduce the student's concern and communicate confidence or encouragement.

Advise

The response should provide a useful suggestion, such as reviewing the assignment or using feedback to improve future work.

The response should perform both functions without contradicting itself.

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b) Important Entities

The important entities that should remain coherent are:

- **Assignment** – the work submitted by the student.
- **Mistakes** – the student's concern about possible errors.
- **You** – the student receiving the response.

The response should clearly refer to these entities.

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c) Three Possible Responses

Response 1

---option "Balanced"
It is completely normal to worry about mistakes after submitting an assignment, so try to stay confident. You can review the assignment and use any feedback you receive to improve your future work.
---option "Encouraging"
Don't worry too much about the mistakes you may have made; submitting your assignment is already an important step. Review the feedback when you receive it, and use it to improve and feel more confident next time.
---option "Supportive"
You have done your part by submitting the assignment, so there is no need to worry excessively. Review any mistakes and feedback later, as they can help you improve and become more confident in your future work.

d) Comparison of the Responses

Criterion	Response 1	Response 2	Response 3
Coherence	Excellent	Excellent	Excellent
Constraint satisfaction	Excellent	Excellent	Excellent
Politeness	Very good	Very good	Very good
Relevance	Very high	Very high	High
Reassurance	Clear	Strong	Clear
Advice	Clear	Clear	Clear

All three responses satisfy the required constraints:

- They contain reassurance.
- They provide advice.
- They maintain the entities assignment, mistakes, and you.
- They use keywords such as **review**, **improve**, **confident**, and **feedback**.
- They contain two sentences.
- They maintain a positive and polite tone.

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e) Best Response and Justification

Response 1 is the best choice.

It provides a balanced combination of reassurance and practical advice. It first addresses the student's worry and then recommends reviewing the assignment and using feedback for improvement.

The response also maintains strong entity coherence because the assignment, possible mistakes, and the student are clearly connected.

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f) Effect of Losing Entity Coherence

Entity coherence means that the system consistently refers to the same entities throughout a conversation.

If entity coherence is not maintained, the response may become confusing or irrelevant.

For example, if the system suddenly talks about **a project** instead of the student's **assignment**, the student may not know whether the response refers to the submitted assignment.

Similarly, if **mistakes** are incorrectly attributed to another person, the meaning of the response changes.

Therefore, maintaining entity coherence is essential for generating relevant, understandable, and trustworthy responses.

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3. Word Sense Disambiguation and Meaning Representation

Source Sentence

“Priya sat on the bank and watched the boats moving across the water.”

The word **bank** has two possible meanings:

1. A financial institution.
2. The side of a river.

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a) Possible Meanings of “Bank”

Meaning 1: Financial Institution

A bank can refer to an organization that provides financial services such as:

- Depositing money.
- Providing loans.
- Managing accounts.

Meaning 2: Side of a River

A bank can also mean the land beside a river or other body of water.

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b) Correct Sense and Justification

The correct meaning is:

Bank = the side of a river.

The contextual evidence strongly supports this interpretation.

The sentence contains:

- **Boats**
- **Water**
- **Moving across the water**
- **Sat on the bank**

A person can sit on the side of a river and watch boats moving across the water.

The financial meaning is semantically incompatible with the rest of the sentence. A person could sit inside a financial bank, but the combination of **boats**, **water**, and **bank** strongly indicates a riverside location.

Therefore:

bank → river bank

This is an example of Word Sense Disambiguation using contextual information.

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c) Predicate Logic Representation

The important entities are:

- **Priya** – person.
- **Bank** – riverside location.
- **Boats** – objects being watched.
- **Water** – body of water.

The important actions and relationships are:

- Priya sits on the bank.
- The bank is beside the water.
- Priya watches the boats.
- The boats move.
- The boats move across the water.

A suitable predicate representation is:

sit(Priya, Bank)

bank(Bank)

beside(Bank, Water)

watch(Priya, Boats)

boat(Boats)

move(Boats)

water(Water)

A more complete representation of the semantic relationships is:

sit(Priya, Bank) ∧ bank(Bank) ∧ beside(Bank, Water) ∧ watch(Priya, Boats) ∧ boat(Boats) ∧ move(Boats) ∧ water(Water)

This representation preserves the important relationships between the person, location, boats, and water.

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d) Unambiguous English Paraphrase

A simple paraphrase is:

"Priya sat on the side of the river and watched the boats moving across the water."

This removes the ambiguity of the word **bank** by explicitly stating that it means the **side of the river**.

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e) Importance of Word Sense Disambiguation in Machine Translation

Word Sense Disambiguation is important in Machine Translation because a word with multiple meanings may require different translations depending on context.

For example, **bank** could refer to:

- A financial institution.
- The side of a river.

If a translation system selects the financial meaning when the sentence refers to a river, the translated sentence may become incorrect or meaningless.

Contextual WSD helps the translation system:

1. Identify ambiguous words.
2. Examine surrounding words.
3. Determine the intended meaning.
4. Select the appropriate translation.
5. Preserve the meaning of the original sentence.

For the given sentence, the presence of **boats** and **water** indicates that **bank** means the side of a river.

Therefore, WSD improves Machine Translation by reducing lexical ambiguity and preserving the intended meaning across languages.

Overall Conclusion

The three scenarios demonstrate important challenges in Natural Language Processing.

In **Reference Resolution**, the system must identify which entities pronouns and referring expressions refer to. Gender, number, recency, grammatical role, semantic compatibility, and discourse coherence all contribute to accurate reference resolution.

In **Dialog Act Recognition**, the system must understand the purpose of a user's statement and generate an appropriate response. The response must maintain entity coherence while satisfying requirements such as reassurance, advice, politeness, and relevance.

In **Word Sense Disambiguation**, the system must determine the intended meaning of ambiguous words by examining contextual information. In the sentence about Priya and the bank, the words **boats** and **water** clearly indicate the riverside meaning.

Overall, accurate NLP requires the system to consider not only individual words but also **context, discourse relationships, grammatical structure, semantic compatibility, and the intended meaning of the complete communication**.
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